Bike Sharing Demand Analysis

Final Project | Methods, Findings, Recommendations and Automation

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1. Executive summary

The three supplied datasets were combined into 1,000 hourly observations, containing 58,304 rentals. Registered users accounted for 53,383 rentals (91.6%); casual users accounted for 4,921 (8.4%). The busiest hour by mean demand was 17:00. Results describe the uploaded historical sample and should not be generalized to other seasons without more data.

2. Objectives and source data

The project brief requests merging, cleaning, statistical analysis, interactive reporting, forecasting, anomaly detection, and repeatable automation. Dataset 1 has 610 partial hourly rows, dataset 2 has 610 matching rows with additional measures, and dataset 3 has 390 complete hourly rows. Dataset 1 and 2 are joined by instant; dataset 3 is appended.

3. Data preparation and quality

- Normalize date and numeric fields and preserve source variables.
- Join the first two sources on instant, append the third, and sort by instant.
- Classify weekend/weekday, season, weather and hour group.
- Check for duplicate IDs, missing values, and cnt = casual + registered.
- Observed checks: 1,000 merged rows, zero duplicate instant IDs, 11 missing cells, and zero arithmetic count inconsistencies. Missing values require review before operational use.

4. Descriptive findings

Measure	Observed value
Hourly observations	1,000
Total rentals	58,304
Registered rentals	53,383 (91.6%)
Casual rentals	4,921 (8.4%)
Mean rentals per observed hour	58.30
Highest mean-demand hour	17:00
Clear-weather average	61.67 rentals/hour
Cloudy/mist average	58.60 rentals/hour

Light rain/snow average	33.20 rentals/hour

Weather comparisons are observational associations, not causal estimates. The heavy rain/snow category has sparse observations and should not drive conclusions on its own.

5. Reporting architecture

Executive Dashboard provides headline KPIs and charts. Clean Data contains the merged source table. Daily Trends aggregates daily totals and flags unusual days. Demand Segments summarizes hours, weather, day type, user type, season, and month. Insights & Forecast contains a seven-day baseline forecast and recommendations. Data Quality reports consistency checks. Read Me describes usage and limitations.

6. Forecasting and anomaly methodology

Daily rentals are forecast using simple exponential smoothing with alpha = 0.35, giving a flat seven-day baseline. The displayed high/low bounds are heuristic (forecast ±1.96 times historical daily standard deviation), not calibrated statistical prediction intervals. Anomalies are days whose daily rental z-score magnitude is at least 2. Incomplete days may appear anomalous simply because fewer hours were recorded.

The workbook does not contain a trained machine-learning model or generative AI service. Its automated insights are deterministic and rule-based; labeling them as independently AI-generated would be misleading.

7. Automation and refresh process

A separate VBA module is supplied. After importing it into a macro-enabled Excel copy, RefreshBikeSharingReport recomputes daily trends, segment summaries, anomaly flags, and the seven-day forecast from the current Clean Data sheet. It also refreshes Excel connections and recalculates formulas. The module does not automatically ingest external source files or install itself in the .xlsx workbook.

The original .xlsx remains macro-free for compatibility. A separate Python build script rebuilds the workbook from the three source files. Neither implementation currently provides native pivot slicers, a Power Query pipeline, or a trained ML model.

8. Business recommendations

1. Rebalance bikes and prioritize service readiness before the observed 17:00 peak.
2. Protect capacity for registered users while testing targeted casual-rider promotions.
3. Incorporate weather information in daily staffing and redistribution decisions.
4. Investigate anomalous days after checking for missing hours and events.
5. Validate forecasting accuracy on a holdout period before using predictions for inventory decisions.
6. Extend the analysis to a full annual cycle to test seasonal generalizability.

9. Limitations and future enhancements

Only 1,000 historical hours across approximately six weeks are available. Weather categories are not equally represented, and no external event or location information is supplied. Proposed next steps are Power Query ingestion, native PivotTables/slicers, rolling forecast validation, and optionally a clearly identified machine-learning model with out-of-sample metrics.

10. Deliverables and reproducibility

Files supplied: Bike_Sharing_Final_Automated_Report.xlsx, Bike_Sharing_Automation.bas, build.py, the three source datasets, this summary, and a README. For a fresh rebuild, install the Python dependencies required by build.py and run it in a folder containing the source files. For Excel-only refresh, save a macro-enabled copy, import the .bas file, and run the macro after updating Clean Data.